Proxy

Lazy-load heavy image assets
Season 2 · GoF · Beginner

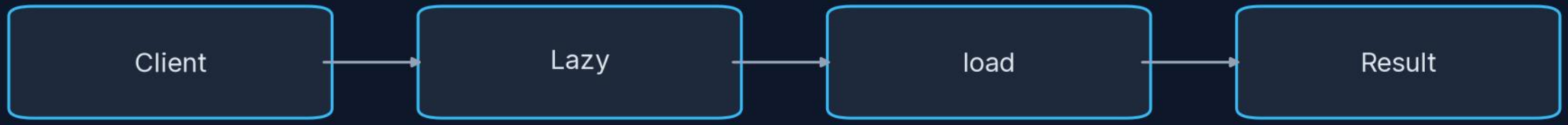
Today's Hook

- Problem: Lazy-load heavy image assets
- Naive pain cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- We will refactor to Proxy with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Lazy-load heavy image assets
- Constraints: solve «Lazy-load heavy image assets» with Proxy — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Lazy-load heavy image assets
- Success looks like: LOAD /big.png

Structure



Proxy: Lazy-load heavy image assets

Real-World Scenario

- Running example: Lazy-load heavy image assets
- Key types: Image, RealImage, LazyImageProxy
- Happy-path stdout cue: LOAD /big.png
- Failure-path stdout cue: Failure path is asserted in ProblemTest (injected fault / invalid input)

Naive Code

```java
package com.buildwithwaqar.patterns.gof.structural.proxy.problem;

import java.util.*;

public final class NaiveImage {
  public NaiveImage(String path){ System.out.println("LOAD "+path); }
  public String display(){ return "shown"; }
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Lazy-load heavy image assets
- Without Proxy, call sites absorb the wrong details.
- Failure mode to show on camera: Failure path is asserted in ProblemTest (injected fault / invalid input)
- New requirements force shotgun edits across the naive code.

Solution

```java
package com.buildwithwaqar.patterns.gof.structural.proxy.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;

interface Image { String display(); boolean loaded(); }
final class RealImage implements Image {
  private final String path; RealImage(String path){ this.path=path; System.out.println("LOAD "+path); }
  public String display(){ return "shown:"+path; } public boolean loaded(){ return true; }
}
final class LazyImageProxy implements Image {
  private final String path; private RealImage real;
  LazyImageProxy(String path){ this.path=path; }
  public String display(){ if(real==null) real=new RealImage(path); return real.display(); }
  public boolean loaded(){ return real!=null; }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.gof.structural.proxy.problem;

import java.util.*;
import java.util.concurrent.*;

public final class ProblemDemo {
  public static void main(String[] args) {
    System.out.println("=== Proxy: lazy-load heavy image ===");
    System.out.println("--- naive (loads immediately) ---");
var n=new NaiveImage("/big.png"); System.out.println(n.display());
System.out.println("--- proxy ---");
Image p=new LazyImageProxy("/big.png");
System.out.println("loadedBefore="+p.loaded());
System.out.println(p.display());
System.out.println("loadedAfter="+p.loaded());
  }
}
```

Walkthrough Output

```
=== Proxy: lazy-load heavy image ===
--- naive (loads immediately) ---
LOAD /big.png
shown
--- proxy ---
loadedBefore=false
LOAD /big.png
shown:/big.png
loadedAfter=true
```

Step-by-Step Trace

- Match stdout to stages — look for: LOAD /big.png
- Then the injected failure: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Re-run: java -cp ... 14-proxy ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```
// Java 25 direction for Proxy
// Prefer records + sealed where they shrink the demo.
// Problem: Lazy-load heavy image assets
```

Trade-offs

- Use Proxy when «Lazy-load heavy image assets» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Proxy in product language.

Related Patterns

- Name the pattern usually confused with Proxy and separate the forces.
- Sketch the roles for «Lazy-load heavy image assets» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Lazy-load heavy image assets
- Naive pain → Proxy boundary → Demo stdout.
- Remember the failure cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Break a ProblemTest assertion, watch it fail, restore it.